

MD. AJIM HOSSAIN

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Motivated chemical engineering student with strong research, data-analysis, and leadership experience, eager to contribute to the scientific and engineering research community.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Chemical Engineering

Dhaka, Bangladesh

Nov 2022 – Present

- CGPA: 3.70/4.00 (6 semesters completed)
- Relevant coursework: Sustainable Process Engineering, Biochemical Engineering, Process Design, Process Safety, Process Control, Heat Transfer, Mass Transfer, Reaction Engineering, Material and Energy Balance

Zonail Degree College

Higher Secondary Certificate (H.S.C.), GPA: 5.00/5.00

February 2022

Saint Louis High School

Secondary School Certificate (S.S.C.), GPA: 5.00/5.00

February 2019

International English Language Testing System (IELTS)

In Progress

RESEARCH EXPERIENCE

Undergraduate Thesis

Jun 2026 – Present

Evaluation of Reusability and Interchangeability of LCA Modules in a Biorefinery Process Repository

Supervisor: Dr. Kawnish Kirtania

- Developing a generalized, matrix-based Life Cycle Assessment (LCA) methodology to evaluate the interchangeability and reusability of modular LCA components

Machine Learning-based Screening of Ionic Liquids for CO₂ Capture

Feb 2025 – Present

Lead Researcher

- Built a ChemBERTa-XGBoost ensemble to predict CO₂ solubility in ionic liquids, benchmarked against Graph Isomorphism Network (GIN) and Graph Attention Network (GAT) baselines
- Applied RDKit-based cheminformatics for molecular representation and screened candidate ionic liquids under near-industrial capture conditions, followed by DFT-based structural feasibility checks in ORCA

Cellulose Oxalate Sorbent for Antibiotic Removal from Wastewater

Feb 2024 – Mar 2025

Undergraduate Researcher

- Synthesized a cellulose oxalate sorbent from waste newspaper through chemical modification, using vacuum filtration for sample processing
- Characterized the sorbent using FTIR and UV-Vis spectroscopy to evaluate its structural and functional suitability for removing antibiotic residues from wastewater

INTERNSHIP EXPERIENCE

Industrial Internship

6 – 20 June 2026

Eastern Refinery Limited (ERL)

Chattogram, Bangladesh

- Gained hands-on exposure to plant-wide operations at ERL, Bangladesh's only petroleum refinery and a subsidiary of the Bangladesh Petroleum Corporation (BPC), which processes crude oil into about 16 petroleum products including LPG, naphtha, kerosene, diesel, and bitumen
- Observed the crude distillation unit and adsorption unit, and learned how crude oil is separated into fractional products across interconnected process units and subunits, while studying the plant's administrative hierarchy and chain of command
- Gained practical exposure to occupational health and safety practices, including emergency and hazard response protocols, and observed professional conduct expected in an industrial plant environment

PUBLICATIONS

A Two-Stage Machine Learning-based Framework for Generating and Validating Novel Ionic Liquids for CO₂ Capture

Accepted (Oral), the 8th International Conference on Chemical Engineering (ICChE 2026)

Dec 17–19, 2026

Department of Chemical Engineering, BUET, Dhaka-1000, Bangladesh — Lead Author

- Developed a ChemBERTa-XGBoost ensemble model to predict CO₂ solubility in ionic liquids, benchmarked against Graph Isomorphism Network (GIN) and Graph Attention Network (GAT) baselines
- Applied RDKit-based cheminformatics for molecular representation and validation of generated ionic liquid candidates
- Modeled and simulated the novel ionic liquid structure: generated 2D structures/properties from SMILES via RDKit, derived 3D coordinates using MDS and Kabsch alignment for rotational equivariance, then ran DFT optimization in ORCA (MPI-parallelized) to assess structural feasibility

TECHNICAL SKILLS

- Programming: Python, MATLAB
- Machine Learning & Cheminformatics: RDKit, ChemBERTa, XGBoost, Graph Neural Networks (GIN, GAT)
- Data Analysis Tools: SPSS, Origin
- Specialized Software: OpenLCA, DWSIM, Aspen HYSYS
- Research Instrumentation: UV Spectrometer, FTIR
- Office Tools: Microsoft Excel, Word, PowerPoint

PROJECT EXPERIENCE

ChE 308: Analysis of a 100 MTPD Linear Alkylbenzene Sulfonic Acid (LABSA) Production Plant

Process Calculation, Economic Evaluation, and Safety Considerations

- Performed detailed process calculations for a 100 MTPD LABSA production plant, including material and energy balances
- Conducted economic evaluation of the plant and assessed key process safety considerations

ChE 306: Simulation and Optimization for Refinery Heat Exchanger Networks

- Simulated the crude oil topping section of Eastern Refinery Limited using Aspen HYSYS
- Optimized the heat exchanger network configuration using HYSYS's built-in optimizer to improve energy efficiency

ChE 308: Machine Learning Prediction of CO₂ Solubility in Ionic Liquids Using Thermodynamic and Molecular Descriptors

- Curated and cleaned a dataset of ionic liquid CO₂ solubility measurements using RDKit and standard Python libraries
- Selected and trained candidate machine learning models in MATLAB to predict CO₂ solubility from thermodynamic and molecular descriptors

ChE 402: Identification of Base Oil Type in an Engine Oil Sample - Mineral, Semi-Synthetic, or Synthetic? (Ongoing)

LEADERSHIP EXPERIENCE

Regional Chair

Aug 2025 – Present

Southern Asia Regional Subcommittee, AIChE

- Leading 6 regional liaisons to strengthen communication across 60+ AIChE student chapters in the Southern Asia region

Regional Liaison

Aug 2024 – Aug 2025

Southern Asia Regional Subcommittee, AIChE

- Facilitated communication between 14 chapters across the region and developed strategies to enhance regional engagement and collaboration

Organizing Secretary

Aug 2026 – Present

BUET AIChE Student Chapter

- Leading the executives of the Events Management Team, alongside the Chair, in organizing events such as webinars, seminars, and orientations

Co-Chair, Events Management Team

May 2025 – Aug 2026

BUET AIChE Student Chapter

- Made decisions regarding event arrangements and content, and coordinated with other teams within the BUET AIChE Student Chapter to ensure the successful running of chapter activities

Executive, Events Management Team

Feb 2024 – May 2025

BUET AIChE Student Chapter

- Organized and managed chapter events, served as an invigilator to foster student engagement, and coordinated with team members to ensure smooth execution of chapter activities

TEACHING & MENTORING EXPERIENCE

Private Tutor

more than 4 Years

Chemistry, Physics, and Mathematics

- Tutored around 15 students in basic chemistry, physics, and mathematics
- Provided consistent academic mentorship, exam preparation, and doubt-clearing support over an extended period

AWARDS AND SCHOLARSHIPS

- Dean's List Award, two academic years (awarded for maintaining a CGPA of 3.75 or higher)
- Merit List Award, five semesters (awarded for ranking in the top 10% of the class each term)
- Technical Scholarship, six semesters (awarded for satisfactory academic results)

ADDITIONAL INFORMATION

- Languages: Bengali (native), English (fluent), Hindi (conversational)
- Volunteering: Organized Chemical Day (2023, 2023–24, 2024); coordinated 5 volunteers at the US Edu-Fair (2024); assisted with the arrangement of Environment Day (2023)